5.2 Identification of Common Adulterants

Aim:

To identify the adulterant in the foods

Apparatus:

Adulterated foods and test tubes.

Procedure:

Definition:

Adulteration is defined as the process by which the quality or the nature of a given substance is reduced through the addition of a foreign or an inferior substance and the removal of a vital element.

Types of adulterants

1. Intentional
2. Incidental

Intentional: Intentional adulterants are those substances that are added as a deliberate act on the part of the adulterer with the intention to increase the margin of profit. E.g. sand, marble chips, stones, mud, chalk powder, water and dyes. These adulterants cause harmful effects on the body.

Incidental: These adulterants are found in food substances due to ignorance, negligence or lack of proper facilities. It is not a willful act on the part of the adulterer. E.g. Pesticides, droppings of rodents, larvae in food.

Table 5.3 Intentional Adulterants and Methods of Detection

S. No	Name of the food	Adulterant	Simple method for detection of adulterant
1	Ghee or butter	Vanaspathi	Take about one teaspoonful of melted ghee or butter with equal quantity of concentrated hydrochloric acid in a test tube and add to it a pinch of cane sugar. Shake well for one minute and test it after 5 minutes. Appearance of crimson colour in lower layer shows the presence of vanaspathi
2.	Vegetable oil	Argemone oil	Add 5ml of concentrated nitric acid to 5ml of sample. Shake carefully, allow to separate. Yellow, orange yellow, crimson colour in the lower acid layer indicates adulteration.
3.	Honey	Molasses (sugar and water)	A cotton wick dipped in pure honey when lighted with a match stick burns. If adulterated the presence of water will not allow the honey to burn. If it is adulterated, it will produce a crackling sound
4.	Rava	Iron particles	By moving a magnet through rava iron particles can be separated.

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